

CCA103 SAD

An Overview of System Analysis and Design

Chapter 1

Chapter 1 Outline

- Part I: Software Development and System Analysis and Design
- Part II: System Development Life Cycle
- Part III: Iterative Development

Learning Objectives

By the end of this topic, you should be able to:

1. Explain the role played in IS development by the systems analyst.
2. Describe the purpose of systems analysis and design when developing information systems.
3. Describe the fundamental systems development life cycle and its four phases.



Part I

Software Development, and System Analysis and Design

- **SDLC** is the **process** of **planning**, **designing**, **building**, and **delivering** an information system to support business needs.
- Developing an **effective information system** is **complex** and much harder than creating small software applications.
- Many IT projects fail because successful system development requires many factors to be handled correctly.
- The **systems analyst** plays a key role by **understanding business problems**, **identifying improvements**, and **designing solutions**.
- The main goal of system development is to **create organizational value**, not just to build a technically impressive system.

Dispute with the
unionized users

CFO does not care
about the user perspectives

Outdated
legacy
application

Poor planning

Organizational issues

Hard-to-use software

Complicated
technical infrastructure

Poorly
designed

Users did
not want the system

Lack of collaboration between system developers
and business users

The Systems Analyst

- The systems analyst plays an important role in information systems development projects.
- The systems analyst works closely with project team members to ensure the right system is developed effectively.
- The systems analyst must understand how technology can be used to solve business problems.
- The systems analyst may also act as a change agent by identifying improvements, designing solutions, and encouraging users to adopt the system

Systems Analyst Skills

- New information systems introduce change to the organization and its people.
- Leading organizational change requires analysts to understand what to change, how to change it, and how to gain support from others.
- Systems analysts need six key skill categories: **technical**, **business**, **analytical**, **interpersonal**, **management**, and **ethical skills**.
- Analysts must communicate and work effectively with multiple **stakeholders**: end users, managers, programmers, and external vendors.
- Analysts must also act with fairness, honesty, and ethics, especially when handling confidential information.

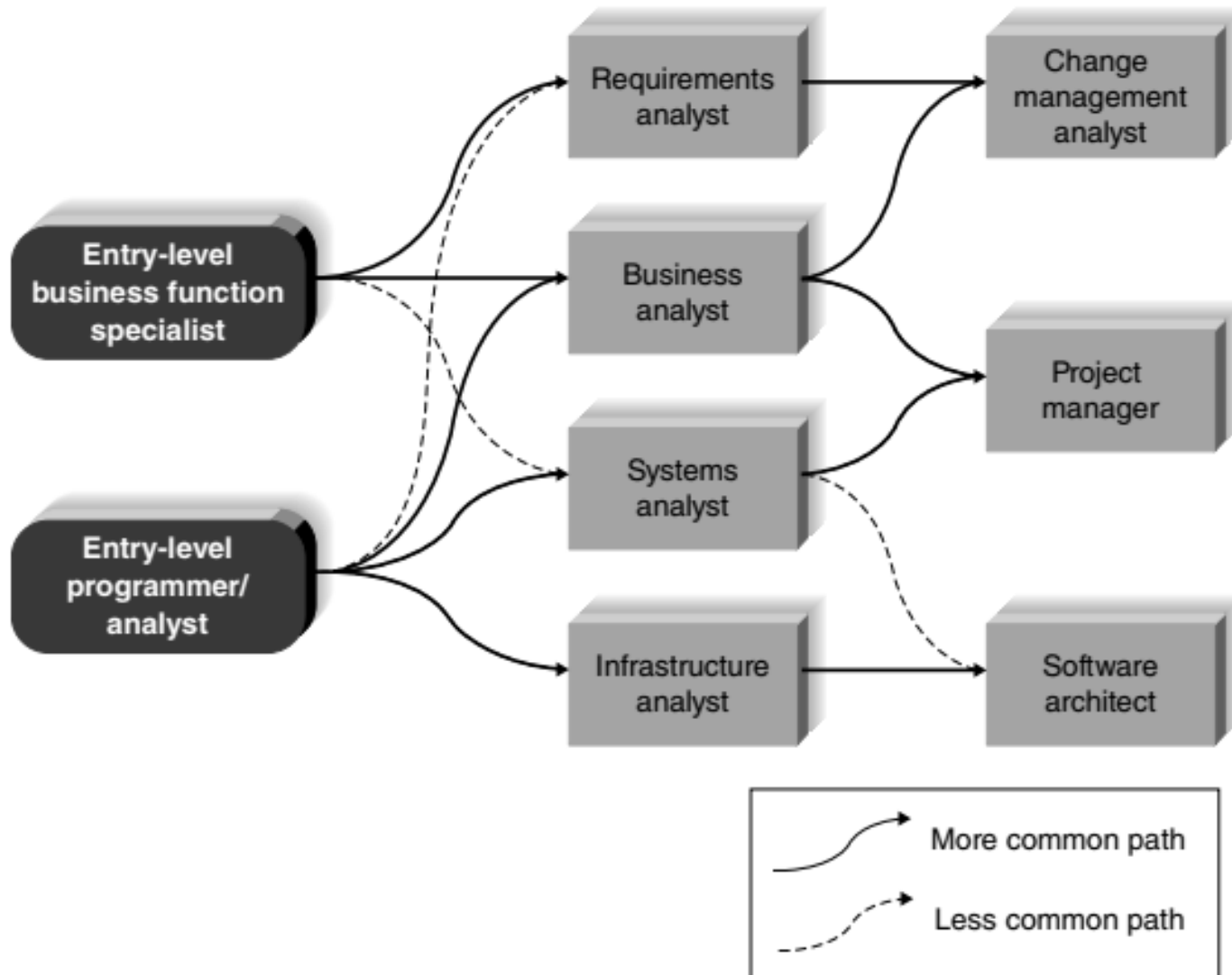
Systems Analyst Roles

- In large organizations, systems development projects involve several analysts with different but complementary roles.
- In smaller organizations, one person may perform multiple analyst roles.
- The **systems analyst** focuses on **information system issues**, including system **design** and **IT-supported business improvement**.
- The **business analyst** focuses on **business value**, business **process** improvement, and **policy design**.
- The **requirements analyst** focuses on gathering **complete** and **accurate requirements** from **stakeholders** using strong business understanding, communication skills, and requirements elicitation techniques.
- The **infrastructure analyst** focuses on **technical infrastructure** issues, ensures the new system aligns with organizational standards, and identifies the hardware, software, network, and database support needed for the system.

Quick activity

Suppose you set a goal to become an analyst after you graduate. What type of analyst would you most prefer to be? Why does this analyst role appeal to you? What type of courses should you take before you graduate? What type of internship should you seek?





Career paths for system developers

Software Development

Systems analysis

Those activities that enable a person to **understand** and **specify** what an **information system** should accomplish

Systems design

Those activities that enable a person to **define** and **describe** in **detail** the **system** that solves the need

Software Development

Computer application (app)

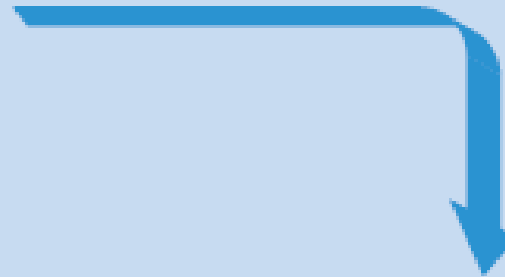
a computer software program that executes on a computing device to carry out a specific set of functions

Information system

a set of interrelated components that collect, process, store, and provide as output the information needed to complete business tasks

Systems analysis

What is required for the new system to solve the problem

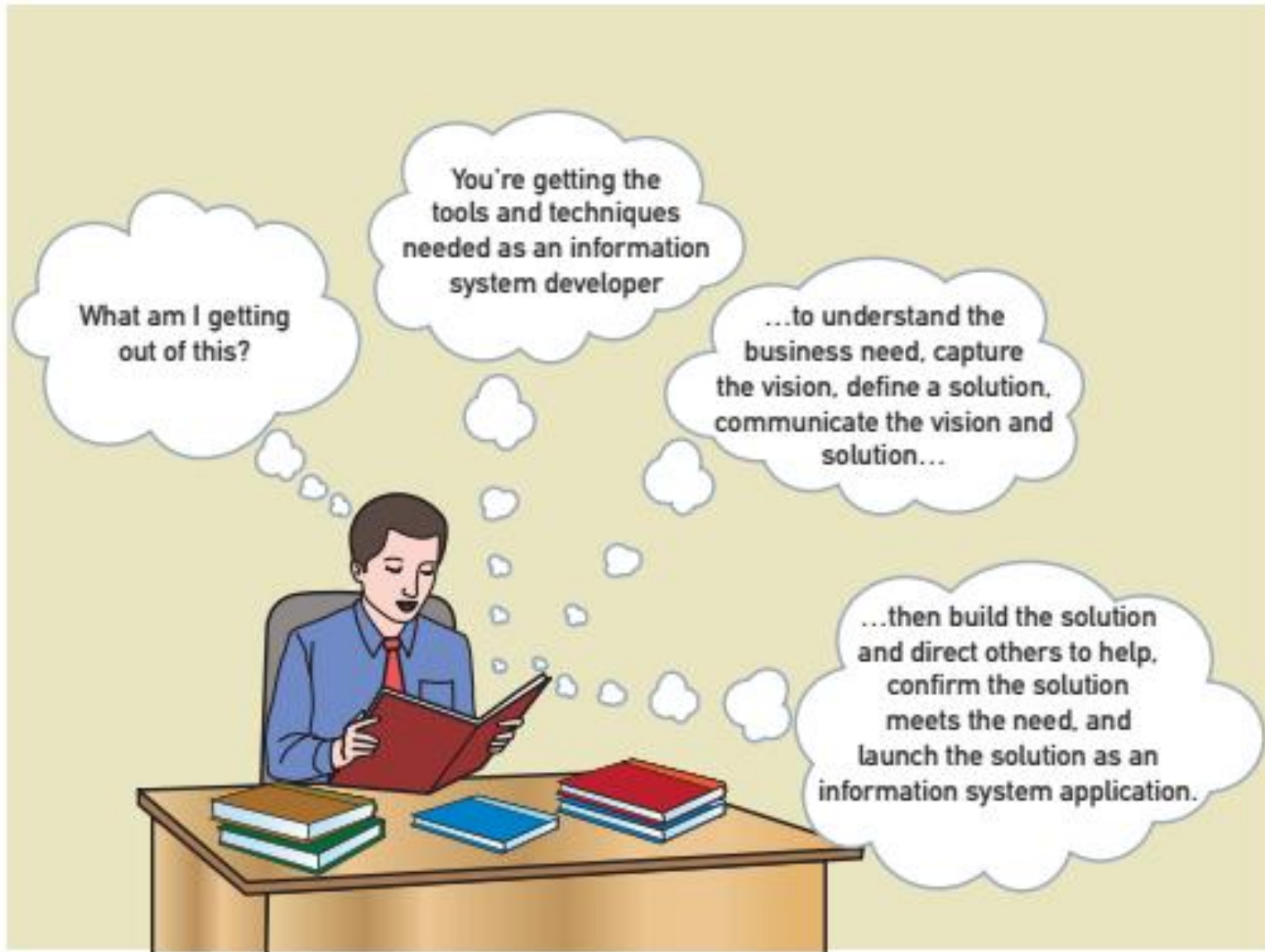


System design

How the system will operate to solve the problem

Software Development





What analysis and design provides for the system developer

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The Systems Development Life Cycle

- Building an information system is like building a house, where ideas are gradually refined before the final product is created.
- The SDLC consists of four main phases: **P**lanning, **A**nalysis, **D**esign, and **I**mplementation.
- Each phase includes a series of steps, techniques, and deliverables that guide system development.
- Although the SDLC may appear linear, projects may follow it **consecutively**, **incrementally**, or **iteratively**.
- The SDLC is a process of gradual refinement, where each phase builds on and adds detail to the previous one.

Planning

The planning phase is the fundamental process of understanding why an IS should be built and determining how the project team will go about building it.

- Step 1: During **project initiation**, the system's business value to the organization is identified
 - how will it contribute to the organization's future success?
 - Most ideas for new systems come from outside the IS area and are recorded on a [system request](#).
- Step 2: The IS approval **committee reviews** the system request and feasibility analysis to decide whether the project should proceed.
 - Once approved, the project moves into project management, where the project manager plans, staffs, and controls the project throughout the SDLC.
 - The main deliverable of this stage is a **project plan** that guides how the system will be developed.

System Request

- A **system request** presents a **summary of a business need** and explains how a system that addresses the need will create **business value**.
- The IS department works with the person or department generating the request (the **project sponsor**) to conduct a feasibility analysis.
- The **feasibility analysis** examines key aspects of the proposed project:
 1. The technical feasibility (Can we build it?)
 2. The economic feasibility (Will it provide business value?)
 3. The organizational feasibility (If we build it, will it be used?)

Elements of the system request form

Element	Description	Examples
Project Sponsor	The person who initiates the project and who serves as the primary point of contact for the project on the business side	Several members of the finance department Vice president of marketing CIO CEO
Business Need	The business-related reason(s) for initiating the system	Reach a new market segment Offer a capability to keep up with competitors Improve access to information Decrease product defects Streamline supply acquisition processes
Business Requirements	The new or enhanced business capabilities that the system will provide	Provide online access to information Capture customer demographic information Include product search capabilities Produce performance reports Enhance online user support
Business Value	The benefits that the system will create for the organization	3% increase in sales 1% increase in market share Reduction in headcount by 5 FTEs \$200,000 cost savings from decreased supply costs \$150,000 savings from removal of outdated technology
Special Issues or Constraints	Issues that pertain to the approval committee's decision	Government-mandated deadline for May 30 System needed in time for the Christmas holiday season Top-level security clearance needed by project team to work with data
FTE, full-time equivalent.		

Analysis

The analysis phase answers the questions of **who** will use the system, **what** the system will do, and **where** and **when** it will be used. During this phase, the project team investigates any current system(s), identifies improvement opportunities, and develops a concept for the new system.

Step 1:

An **analysis strategy** is developed to guide the project team by examining the current system (**as-is system**) and its problems, while identifying ideas for the new system (**to-be system**).

Step 2:

The project team **gathers** and analyzes **requirements** to develop a **system concept**, supported by requirement statements and business analysis models.

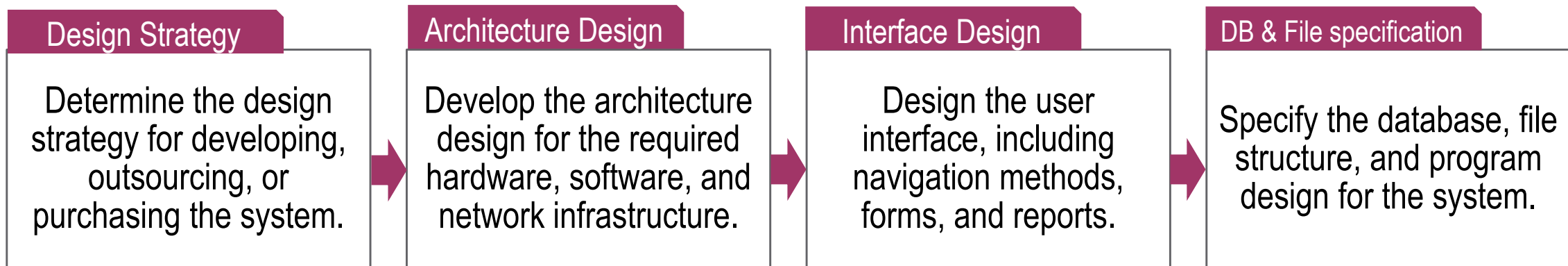
Step 3:

The analyses, system concept, requirements, and models are compiled into a **system proposal**, which is presented to the project sponsor and key decision makers to determine whether the project should proceed.

System Specification

Design

The design phase determines how the system will **operate**, including the **hardware**, **software**, and **network infrastructure** to be in place; the **user interface**, **forms**, and **reports** to be used; and the specific programs, **databases**, and **files** required. Although most strategic decisions about the system are made during the analysis phase when developing the system concept, the steps in the design phase determine exactly how the system will operate.



Implementation

The final phase in the SDLC is the implementation phase, during which the **system is actually built** (or purchased and installed if the design calls for a prewritten software package).

This is the phase that usually gets the most attention, because for most systems it is the longest and most expensive single part of the development process.

System construction

The system is **built** and tested to make sure it works as intended. **Testing** is very important because fixing **errors** later can be costly.

System Installation

The new system has been installed, **replacing the old one**, and users have been **trained** to use it.

Support Plan

A support plan is prepared, including **review** and **future** system **improvements**.

Project Identification and Initiation

- Projects begin with **business needs** that require system support.
- These needs may come from **new strategies, expansion efforts, or organizational change**.
- Some projects are triggered by **business problems** such as poor service, defects, or stronger competition.
- New systems help organizations **improve processes** and **achieve strategic goals**.
- Project ideas may also come from **innovative uses of information technology**.
- Organizations that adopt emerging technologies early can gain a competitive advantage.

Project – a planned undertaking that has a beginning and an end and that produces some definite result

- Used to develop an information system
- Requires knowledge of systems analysis and systems design tools and techniques

System Development Life Cycle (SDLC)

System development process – the actual approach used to develop a particular information system (methodology)

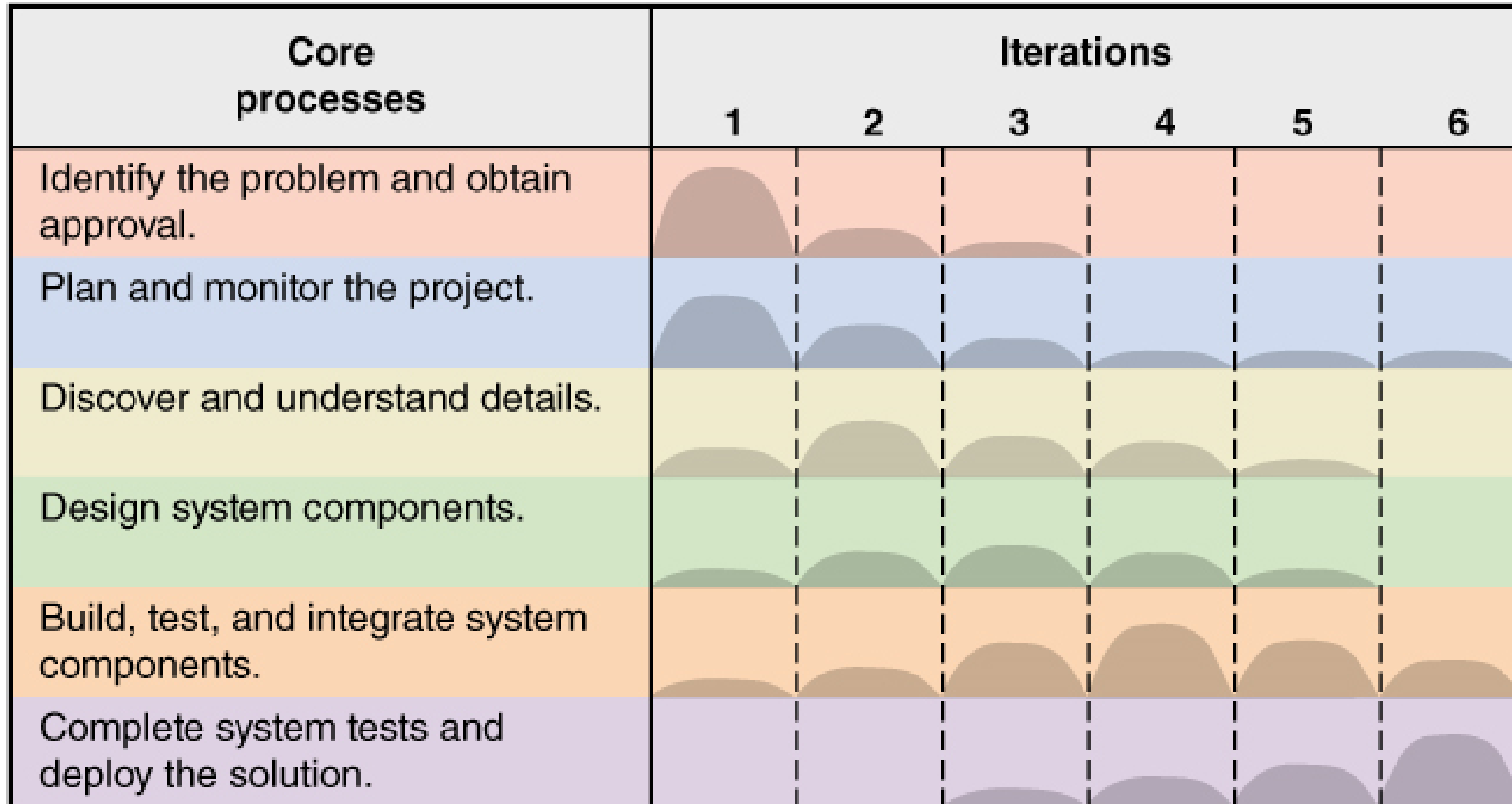
1. Unified process (UP)
2. Extreme Programming (XP)
3. Scrum

Most processes/methodologies now use Agile and Iterative development

Iterative Development

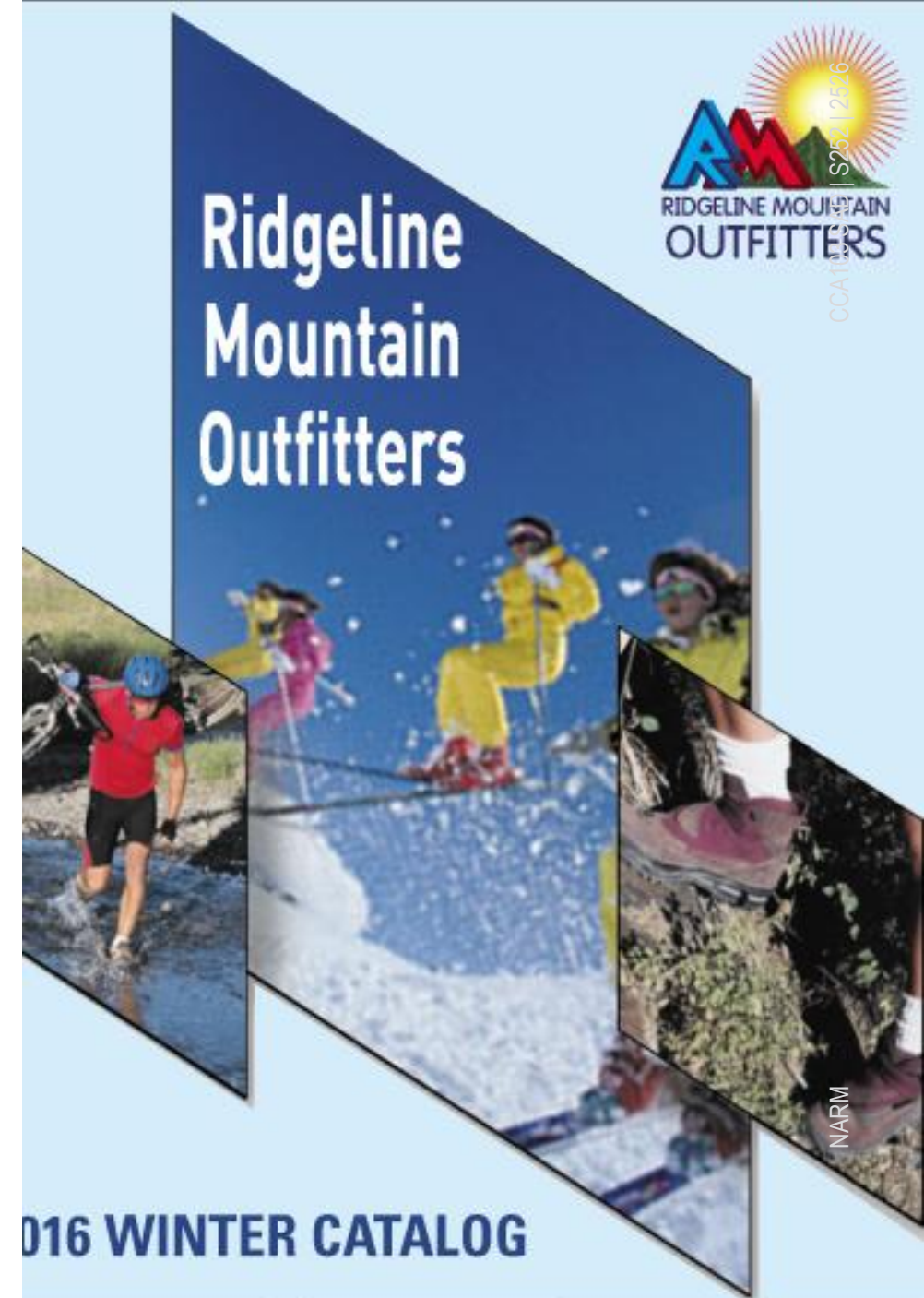
- **Agile development** – an information system development process that emphasizes **flexibility** to anticipate new requirements during development
 - Fast on feet; responsive to change
- **Iterative development** -- an approach to system development in which the system is “**grown**” **piece by piece** through multiple iterations
 - Complete a small part of the system (**mini-project**), then repeat processes to refine and add more, then repeat to refine and add more, until done

Iterative and Agile Systems Development Lifecycle (SDLC)

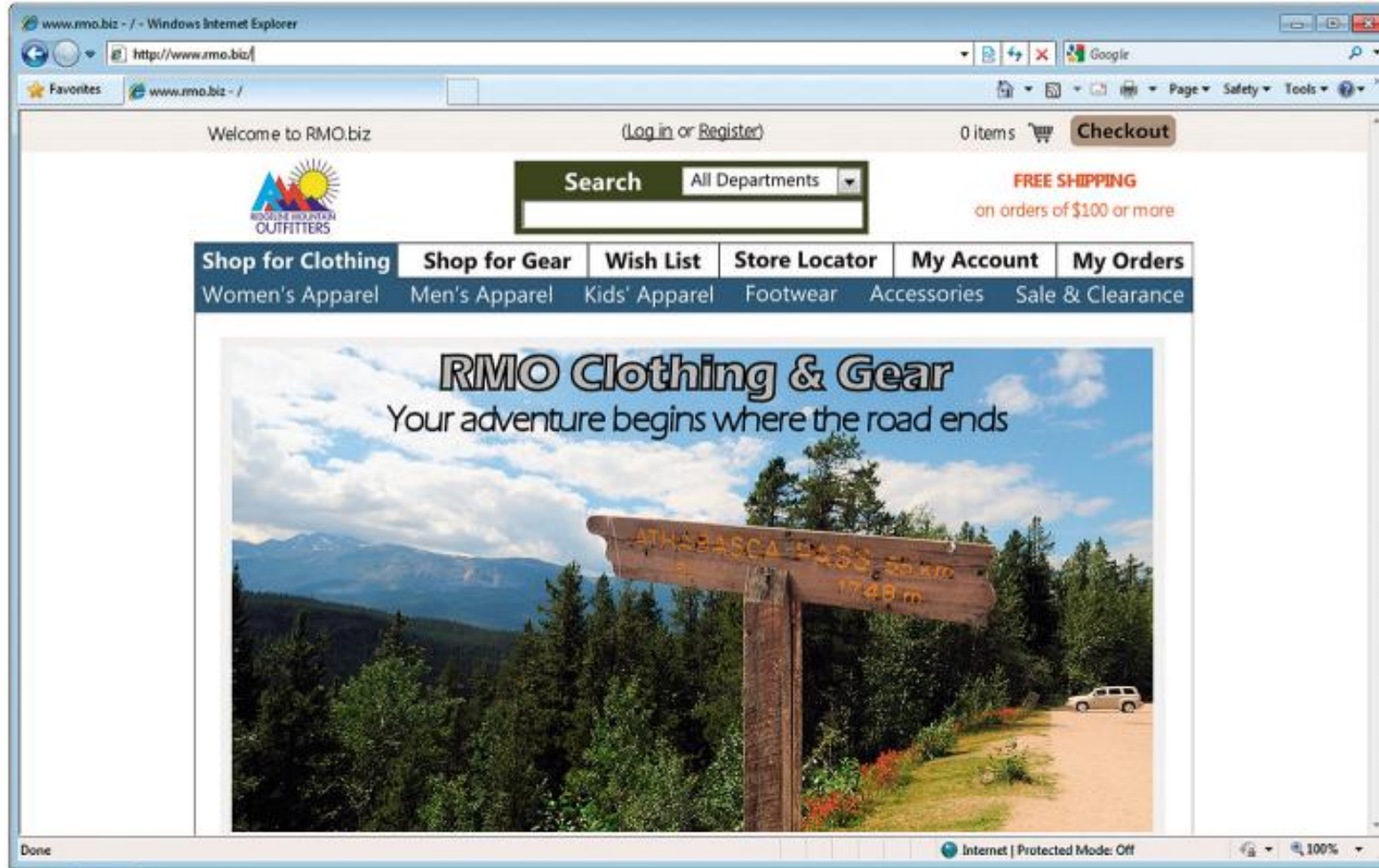


Ridgeline Mountain Outfitters (RMO)

- Large Retail Company
 - Outdoor and sporting clothing and accessories
 - Skiing, mountain biking, water sports
 - Hiking, camping, mountain climbing
- Rocky Mountain and Western States
 - Started mail order and phone order
 - Added retail stores
 - Added extensive E-business component



Ridgeline Mountain Outfitters (RMO) (2 of 2)



RMO Tradeshow System (1 of 2)

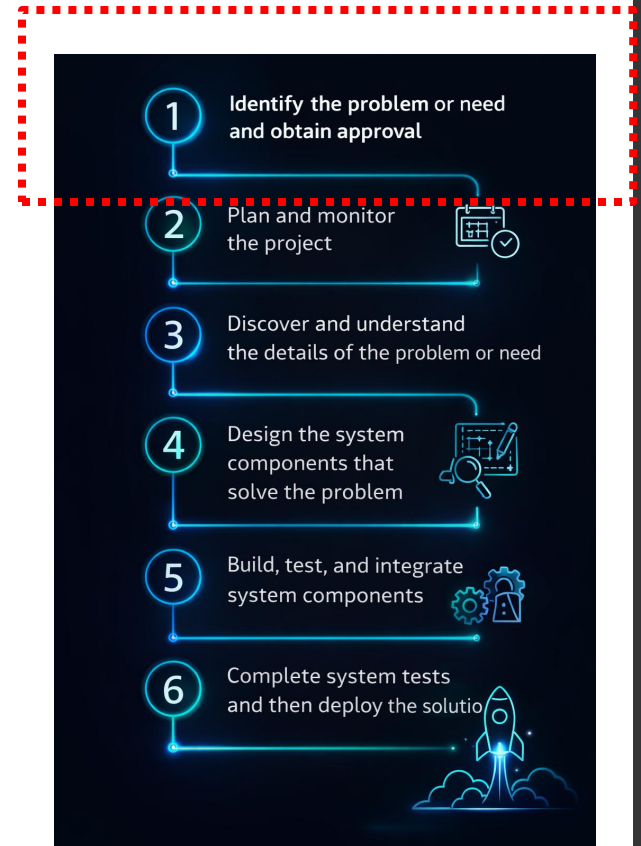
- Sample project for chapter
- Small information system (app)
- Being added to a larger supply chain management system
- Demonstrates one iteration of the small project – assumes more iterations in the total project
- Goes through all six core processes of S D L C
- The plan for this chapter is to complete the iteration in six days

RM0 Tradeshow System (2 of 2)

- **Problem**-- purchasing agents attend apparel and fabric trade shows around the world to order new products from suppliers
- **Need**-- information system (app) to collect and track information about suppliers and new products while at tradeshow
- **Tradeshow Project**-- is proposed
 - Supplier information subsystem (subsystem #1)
 - Product information subsystem (subsystem #2)

D-0: Initial Activities (**pre-project**)

- **Identify** the **problem** and document the **objective** of the system (**core process 1**)
 - Preliminary investigation
 - System Vision Document
- Obtain **approval** to commence the project (core process 1)
 - Meet with key stakeholders, including executive management
 - Decision reached, approve plan and budget



Problem Description

Trade shows have become an important information source for new products, new fashions, and new fabrics. In addition to the large providers of outdoor clothing and fabrics, there are many smaller providers. It is important for RMO to capture information about these suppliers while the trade show is in progress. It is also important to obtain information about specific merchandise products that RMO plans to purchase. Additionally, if quality photographs of the products can be obtained while at the trade show, then the creation of online product pages is greatly facilitated.

It is recommended that a new system be developed and deployed so field purchasing agents can communicate more rapidly with the home office about suppliers and specific products of interest. This system should be deployed on portable equipment.

System Capabilities

The new system should be capable of:

- Collecting and storing information about the manufacturer/wholesaler (suppliers)
- Collecting and storing information about sales representatives and other key personnel for each supplier
- Collecting information about products
- Taking pictures of products (and/or uploading stock images of products)
- Functioning as a stand-alone without connection
- Connecting via Wi-Fi (Internet) and transmitting data
- Connecting via telephone and transmitting data

Business Benefits

It is anticipated that the deployment of this new system will provide the following business benefits to RMO:

- Increase timely communication between trade show attendees and home office, thereby improving the quality and speed of purchase order decisions
- Maintain correct and current information about suppliers and their key personnel, thereby facilitating rapid communication with suppliers
- Maintain correct and rapid information and images about new products, thereby facilitating the development of catalogs and Web pages
- Expedite the placing of purchase orders for new merchandise, thereby catching trends more rapidly and speeding up product availability

System Vision Document

Problem description
System capabilities
Business benefits

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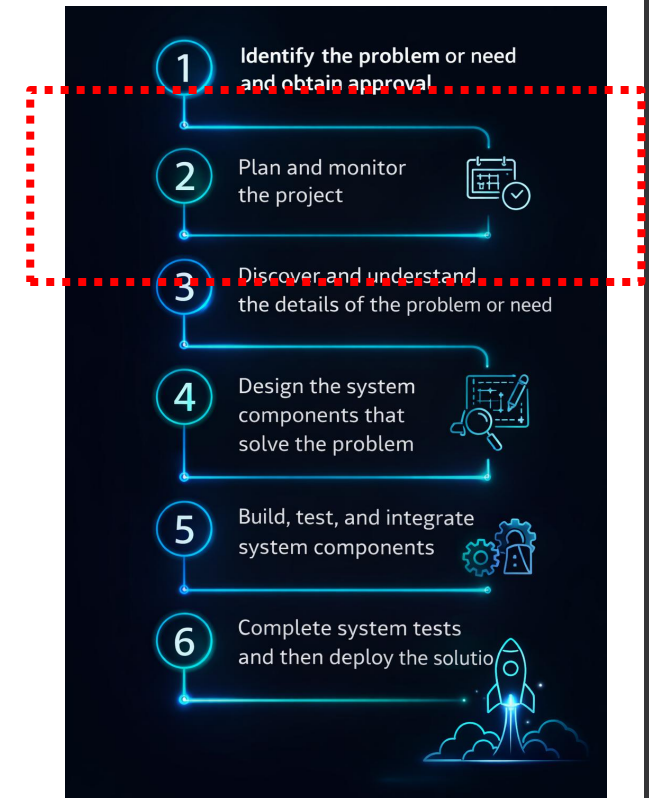
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D-1 Activities

- **Core Process 2: Plan** the Project
 - Determine the major components (functional areas) that are needed
 - Supplier information subsystem
 - Product information subsystem
 - **Define** the **iterations** and assign each function to an iteration
 - Decide to do Supplier subsystem first
 - Plan one iteration as it is small and straightforward
 - **Determine** team **members** and **responsibilities**

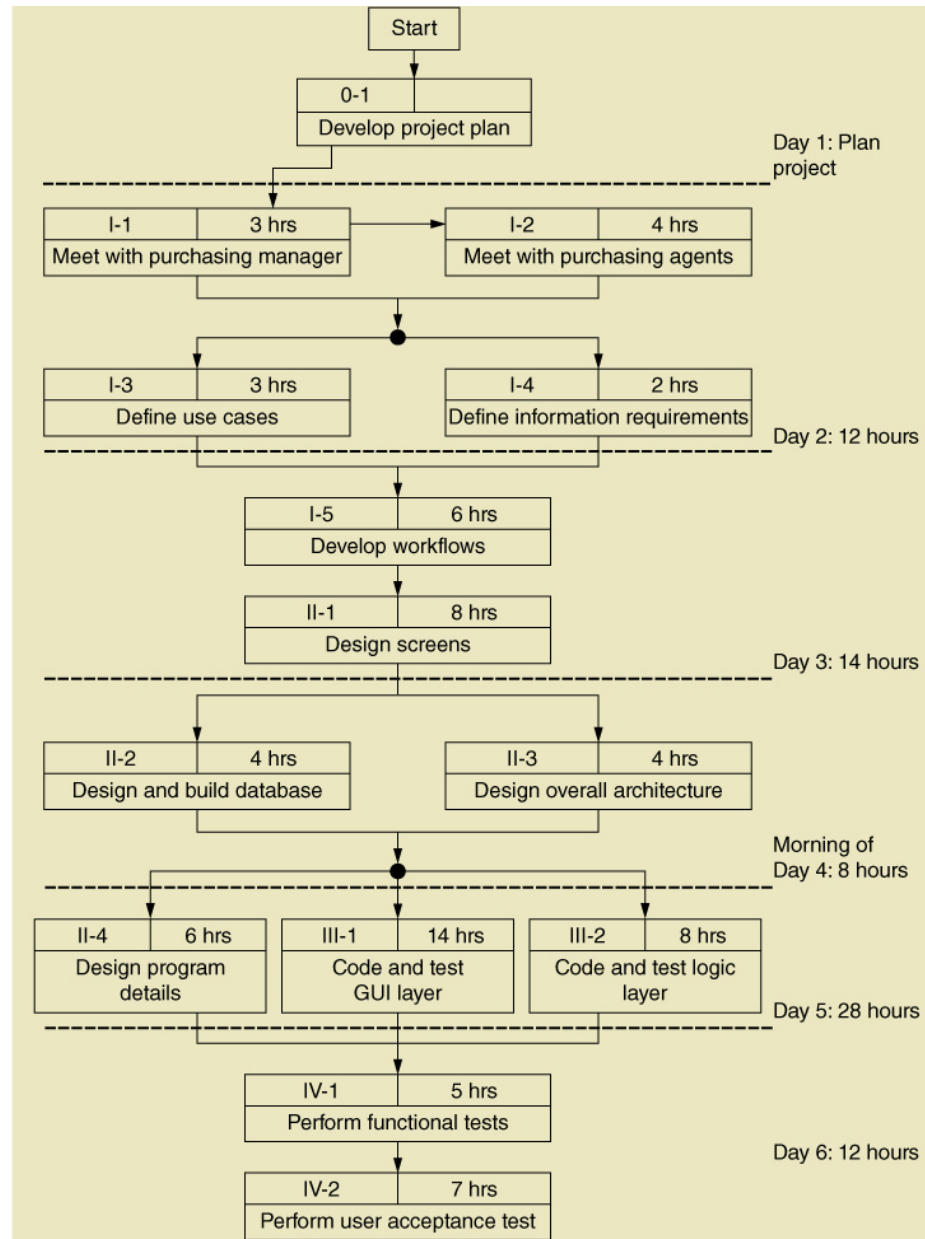


Work Breakdown Structure

- I. Discover and understand the details of all aspects of the problem.
 1. Meet with the Purchasing Department manager. ~ 3 hours
 2. Meet with several purchasing agents. ~ 4 hours
 3. Identify and define use cases. ~ 3 hours
 4. Identify and define information requirements. ~ 2 hours
 5. Develop workflows and descriptions for the use cases. ~ 6 hours
- II. Design the components of the solution to the problem.
 1. Design (lay out) input screens, output screens, and reports. ~ 8 hours
 2. Design and build database (attributes, keys, indexes). ~ 4 hours
 3. Design overall architecture. ~ 4 hours
 4. Design program details. ~ 6 hours
- III. Build the components and integrate everything into the solution.
 1. Code and unit test GUI layer programs. ~ 14 hours
 2. Code and unit test Logic layer programs. ~ 8 hours
- IV. Perform all system-level tests and then deploy the solution.
 1. Perform system functionality tests. ~ 5 hours
 2. Perform user acceptance test. ~ 8 hours

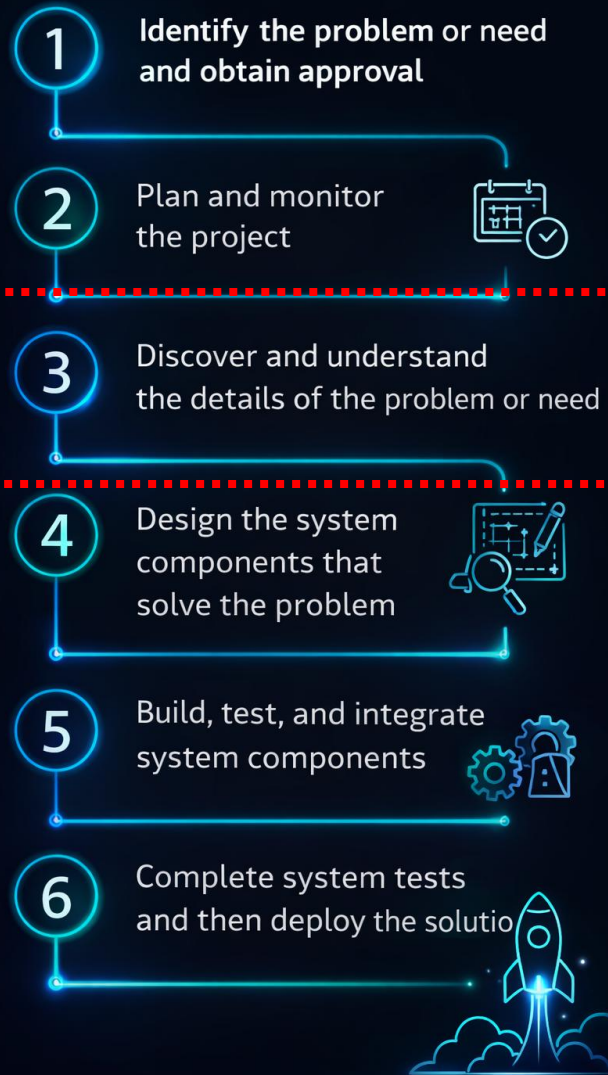
Work Breakdown Structure for Iteration

Describes the work and covers Core Processes 3, 4, 5, and 6



Work Sequence Draft for Iteration

Elaborates the Work Breakdown Structure



D-2: Activities

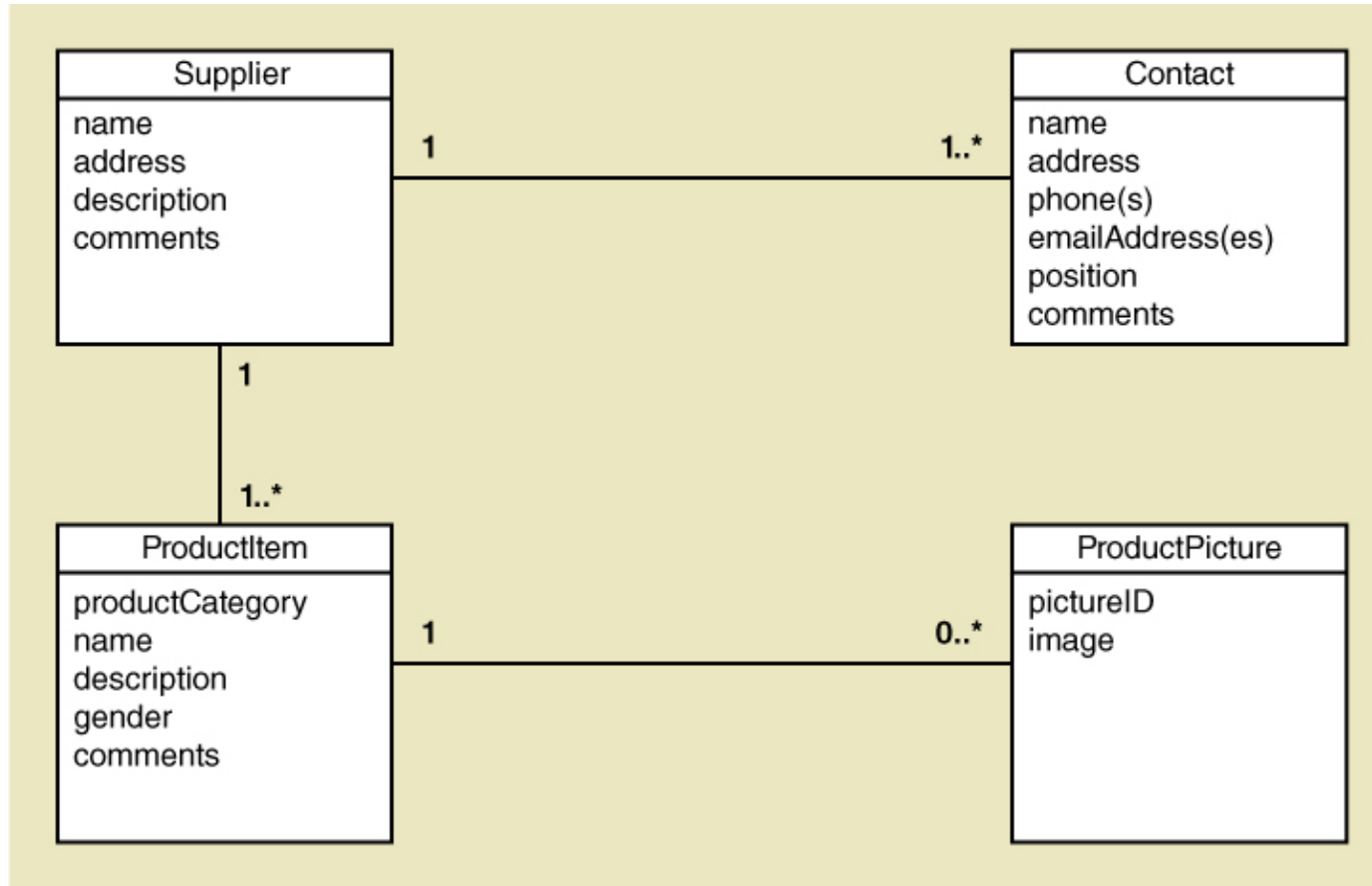
- **Core Process 3: Discover** and Understand Details
 - Do preliminary fact-finding to understand requirements
 - Develop a preliminary list of use cases and a use case diagram
 - Develop a preliminary list of classes and a class diagram

Identify Use Cases: Both subsystems

Use Case	Description
Look up supplier	Using supplier name, find supplier information and contacts
Enter/update supplier information	Enter (new) or update (existing) supplier information
Look up contact	Using contact name, find contact information
Enter/update contact information	Enter (new) or update (existing) contact information
Look up product information	Using description or supplier name, look up product information
Enter/update contact information	Enter (new) or update (existing) product information
Upload product image	Upload images of the merchandise product

Identify Object Classes: Both subsystems

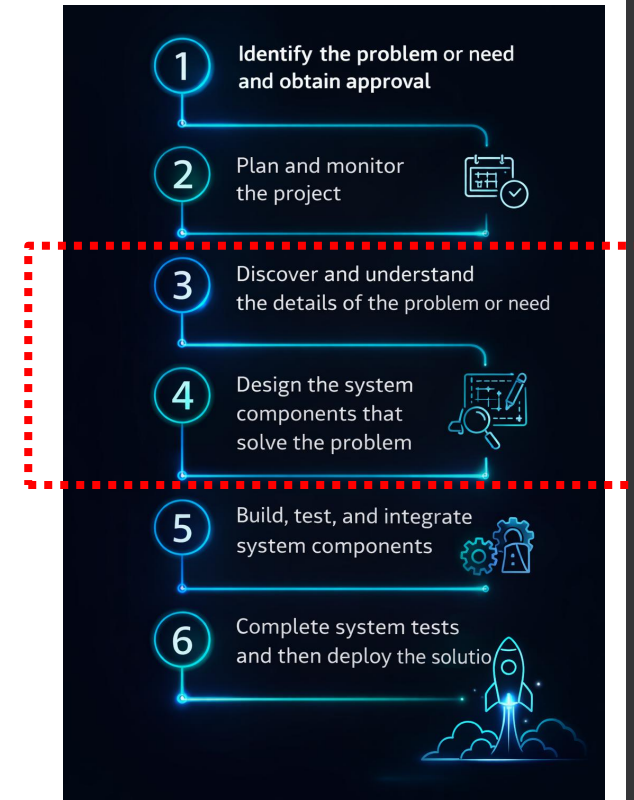
Object Classes	Attributes
Supplier	Supplier name, address, description, comments
Contact	name, address, phone(s), e-mail address(es), position, comments
Product	Category, name, description, gender, comments
ProductPicture	ID, image



Preliminary Class Diagram: Both subsystems

D-3: Activities

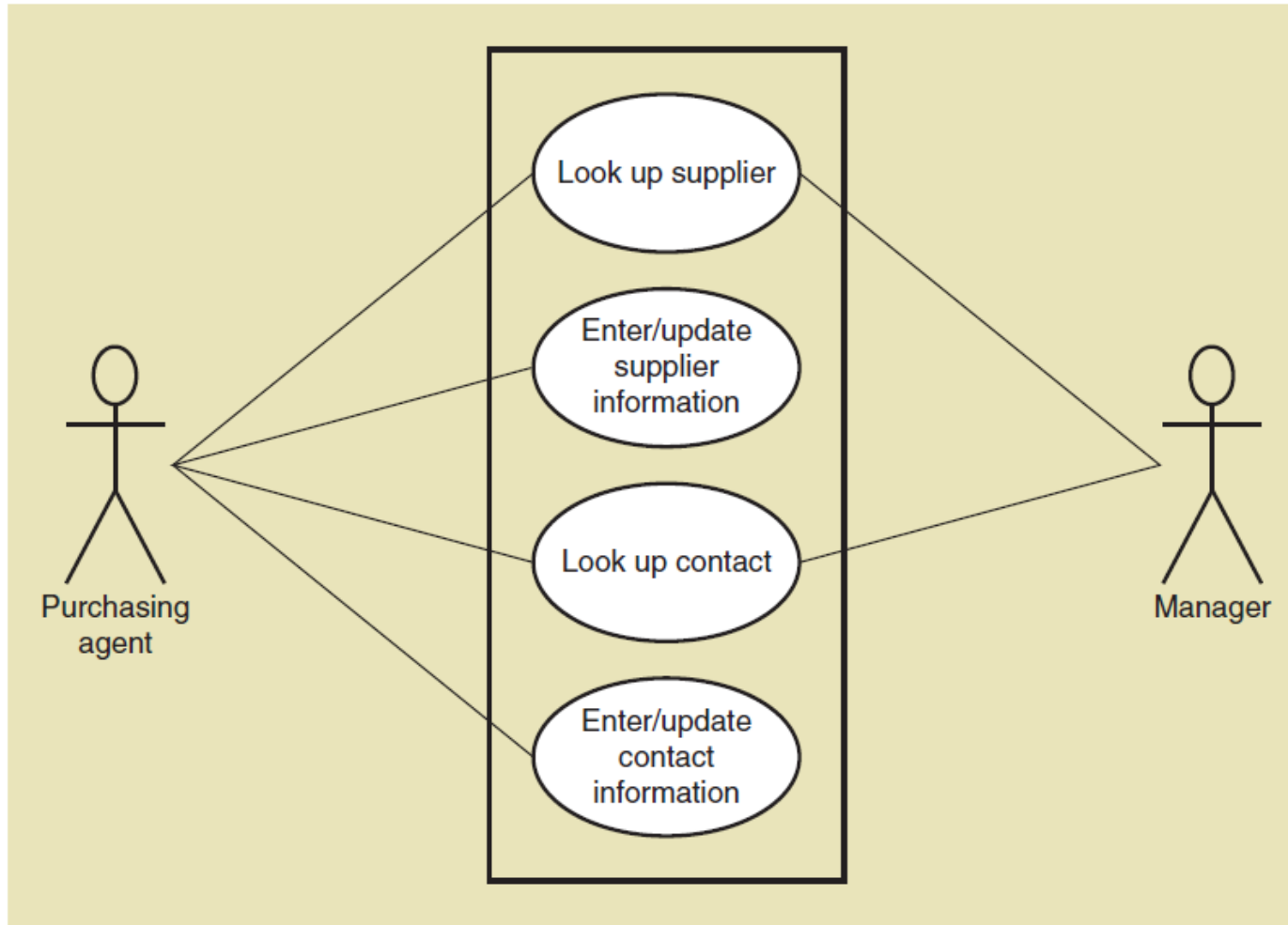
- **Core Process 3: Discover and Understand Details**
 - Do in-depth fact-finding to understand requirements
 - Understand and document the detailed workflow of each use case
- **Core Process 4: Design System Components**
 - Define the user experience with screens and report sketches



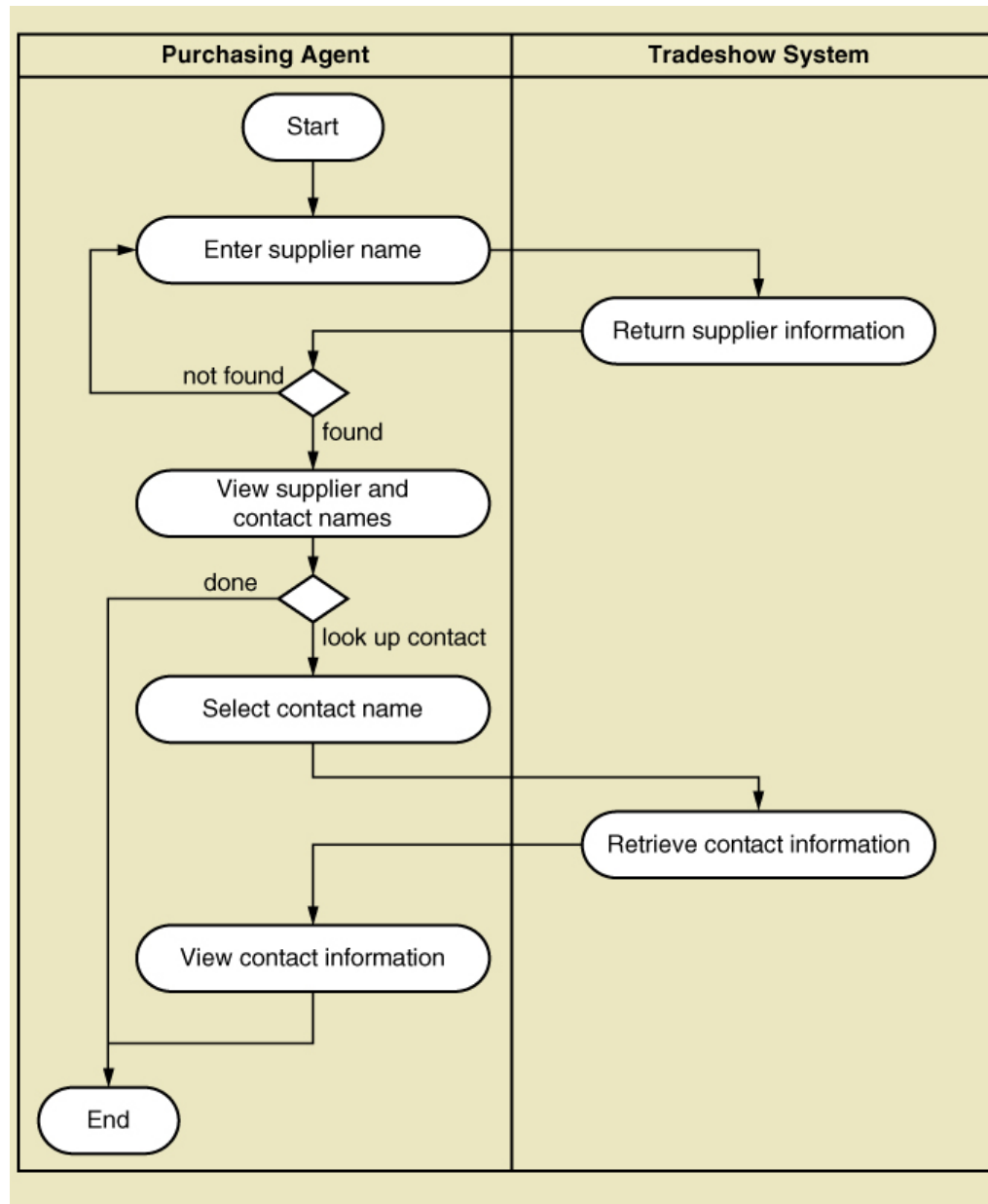
Supplier Information Subsystem

Use cases:

- Look up the supplier
- Enter/update supplier information
- Lookup contact information
- Enter/update contract information



Use Case Diagram: Supplier information subsystem



Activity Diagram (Workflow)

Look up supplier use case

Logo

Web Search

GO

RMO Database Search

Supplier Name

Product Category

Product

Country

Contact Name

GO

Search Results

Supplier Name	Contact Name	Contact Position

Draft Screen Layout: *Look up supplier* use case

D-4: Activities

Core Process 4: Design System Components

- Design the database (schema)
- Design the system's high-level structure
 - Browser, Windows, or smartphone
 - Architectural configuration (components)
 - Design class diagram
 - Subsystem architectural design

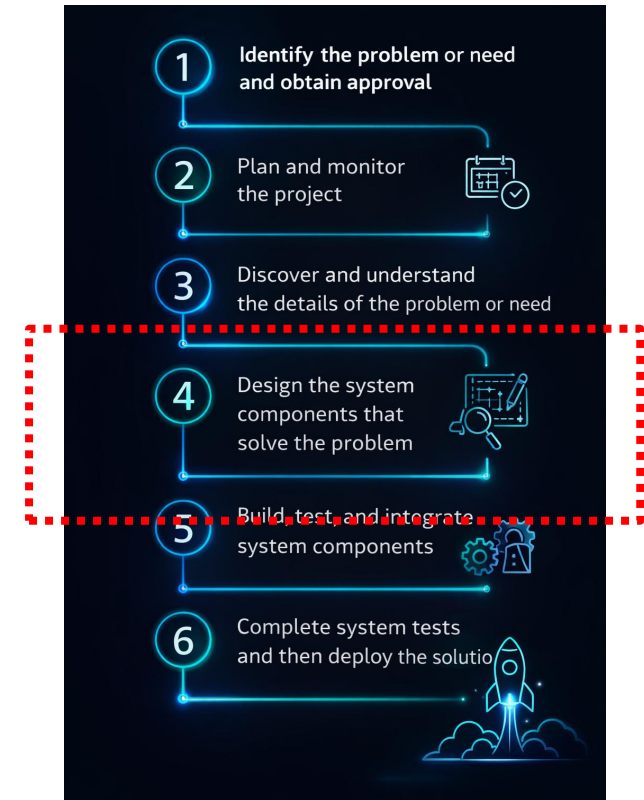
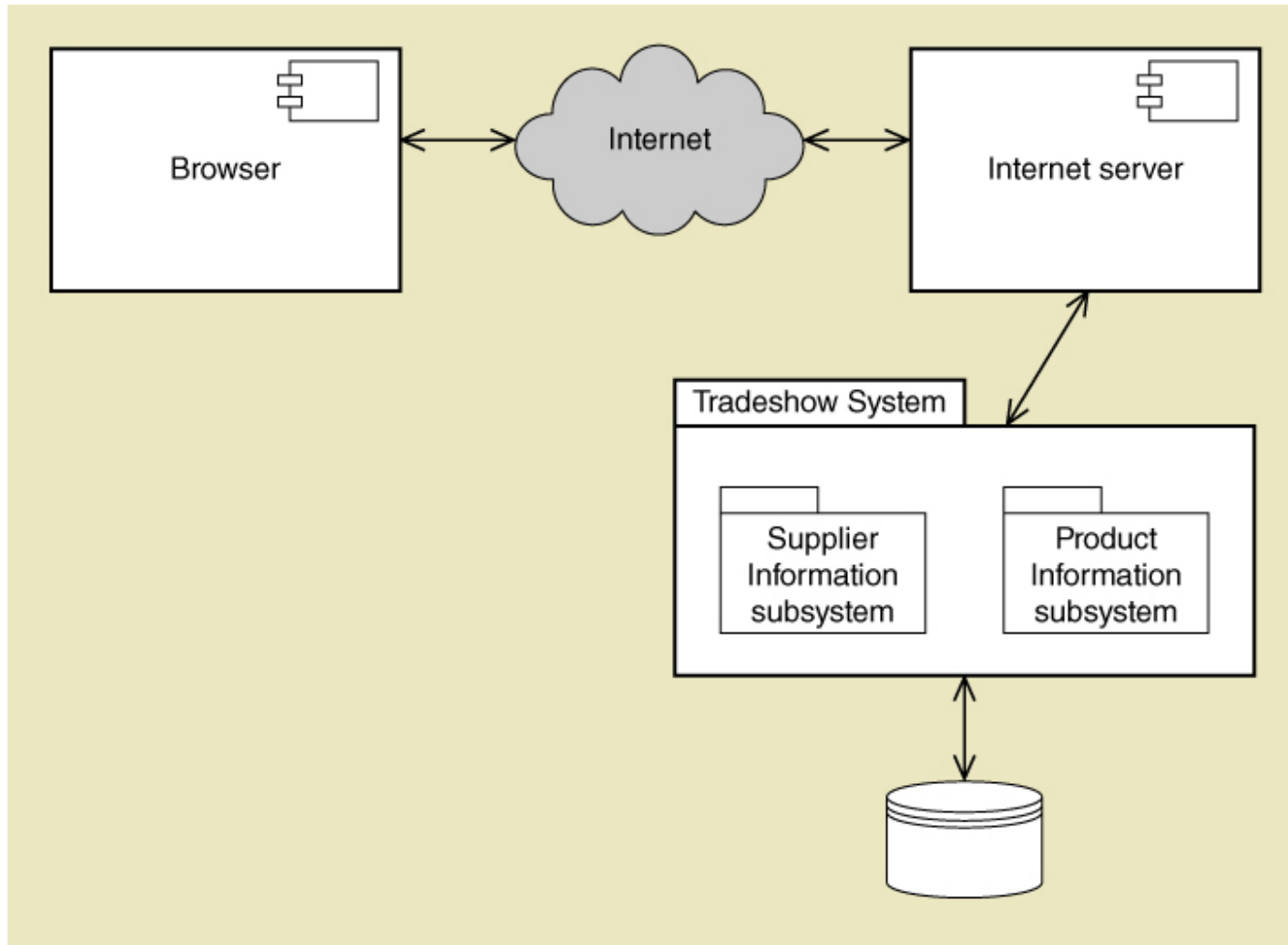


Table name	Attributes
Supplier	SupplierID: integer {key} Name: string {index} Address1: string Address1: string City: string State-province: string Postal-code: string Country: string SupplierWebURL: string Comments: string

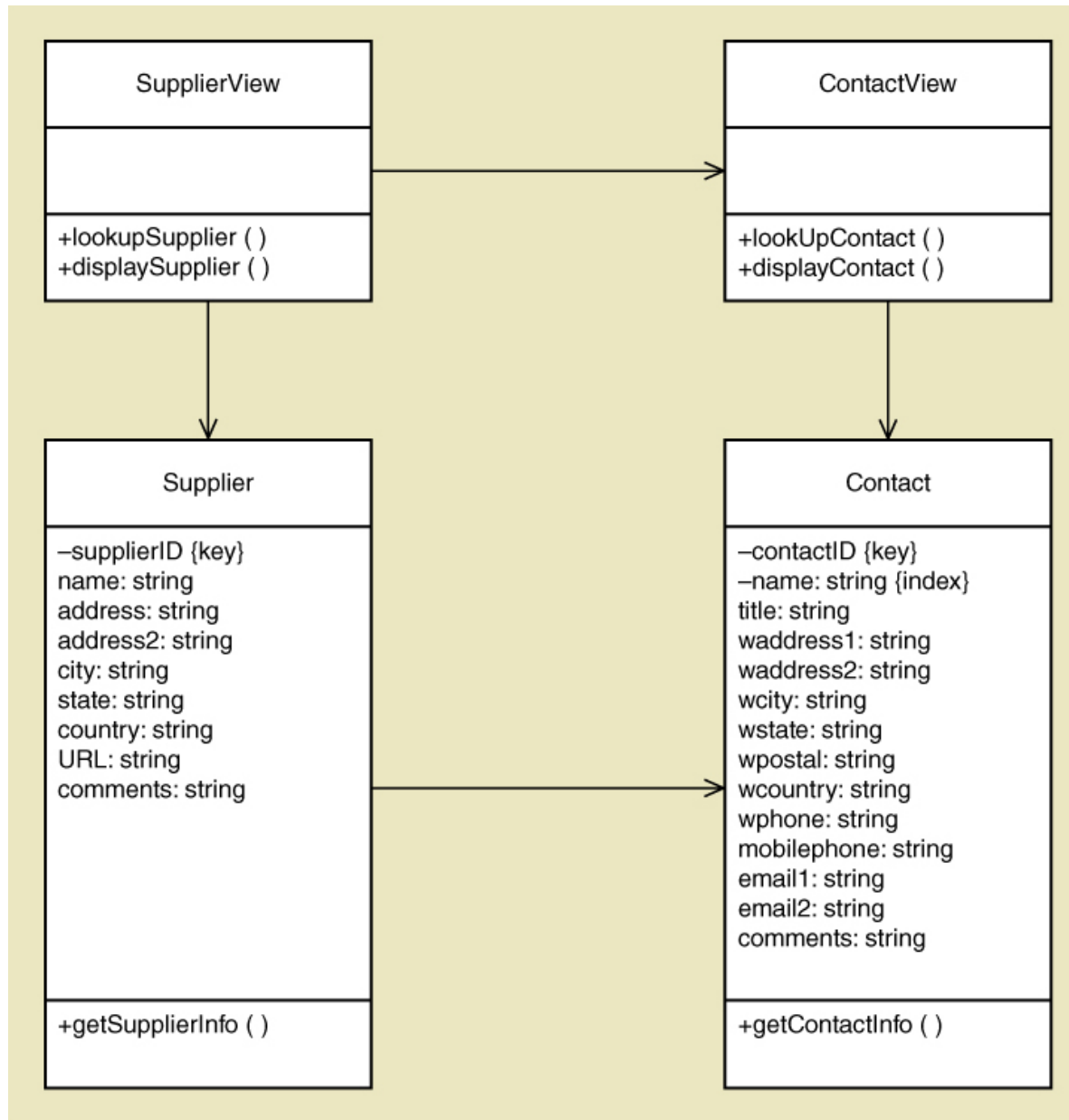
Database Schema (1 of 2)

Table name	Attributes
Contact	ContactID: integer {key} SupplierID: integer {foreign key} Name: string {index} Title: string WorkAddress1: string WorkAddress2: string WorkCity: string WorkState: string WorkPostal-code: string WorkCountry: string WorkPhone: string WorkPhone: string EmailAddress1: string EmailAddress2: string Comments: string

Database Schema (2 of 2)

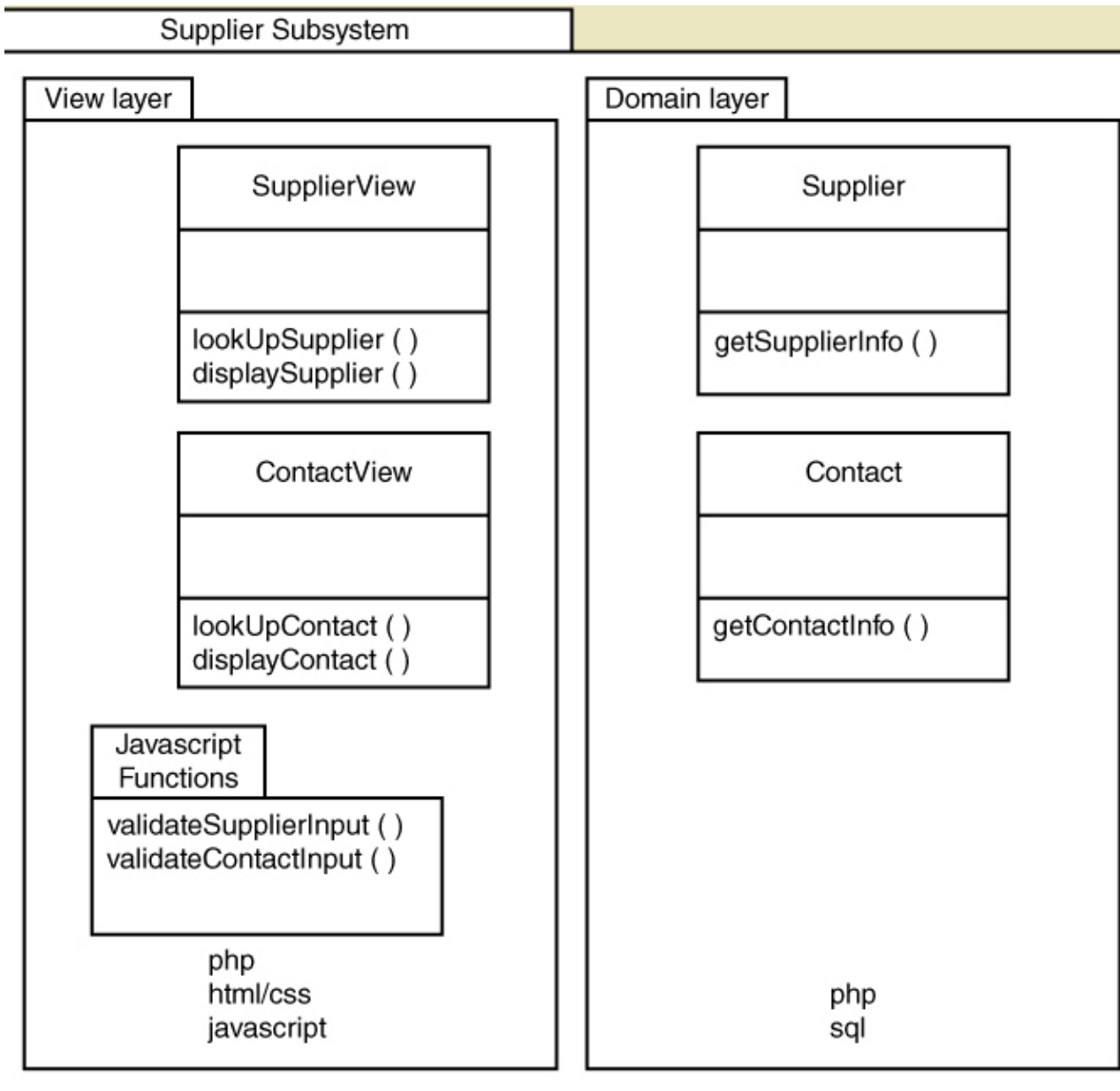


Architectural Configuration Diagram



Preliminary Design Class Diagram

Includes View Layer Classes and Domain Layer Classes



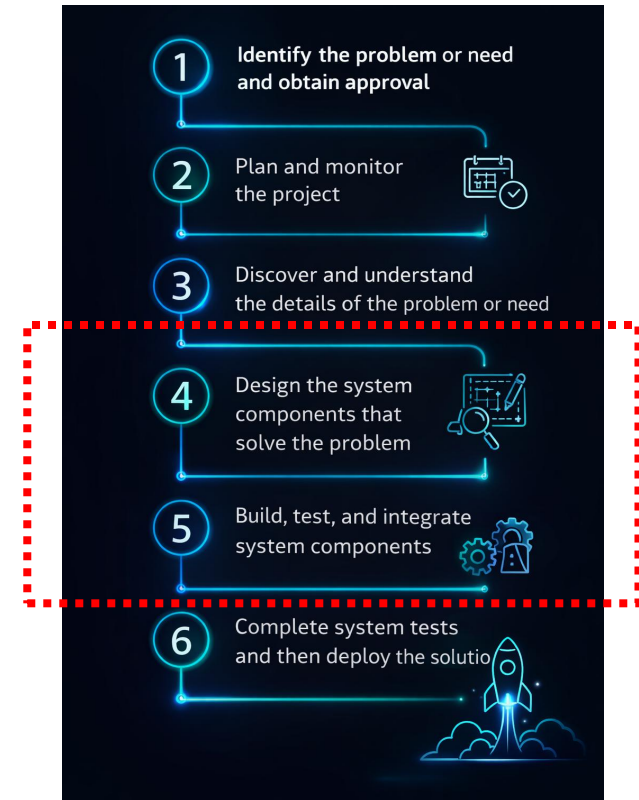
Subsystem Architectural Design Diagram

Notes on Managing the Project

- Lots of design diagrams shown
 - Design in a complex activity with multiple levels
 - One diagram builds on/complements another
 - Not everything is diagrammed, especially for a small project. Pick and choose.
- Programming is also done concurrently
 - You don't design everything, then code
 - You do some design, some coding, some design, some coding

D-5: Activities

- **Core Process 4: Design System Components**
 - Continue with design details
 - Proceed use case by use case
- **Core Process 5: Build, Test, and Integrate System Components**
 - Continue programming (build)
 - Build use case by use case
 - Perform unit and integration tests



```

<?php
class SupplierView
{
    private Supplier $theSupplier;

    function __construct()
    {
        $this->theSupplier = new Supplier();
    }

    function lookupSupplier()
    {
        include('lookupSupplier.inc.html');
    }

    function displaySupplier()
    {
        include('displaySupplierTop.inc.html');
        extract($_REQUEST); // get Form data
        //Call Supplier class to retrieve the data
        $results = $theSupplier->getSupplierInfo($supplier, $category,
                                                $product, $country, $contact);

        foreach ($results as $resultItem){
            ?>
                <tr>
                    <td style="border:1px solid black">
                        <?php echo $resultItem->supplierName?></td>
                    <td style="border:1px solid black">
                        <?php echo $resultItem->contactName?></td>
                    <td style="border:1px solid black">
                        <?php echo $resultItem->contactPosition?></td>
                </tr>
            <?php }
            include('displaySupplierFoot.inc.html');
        }
    }
?>

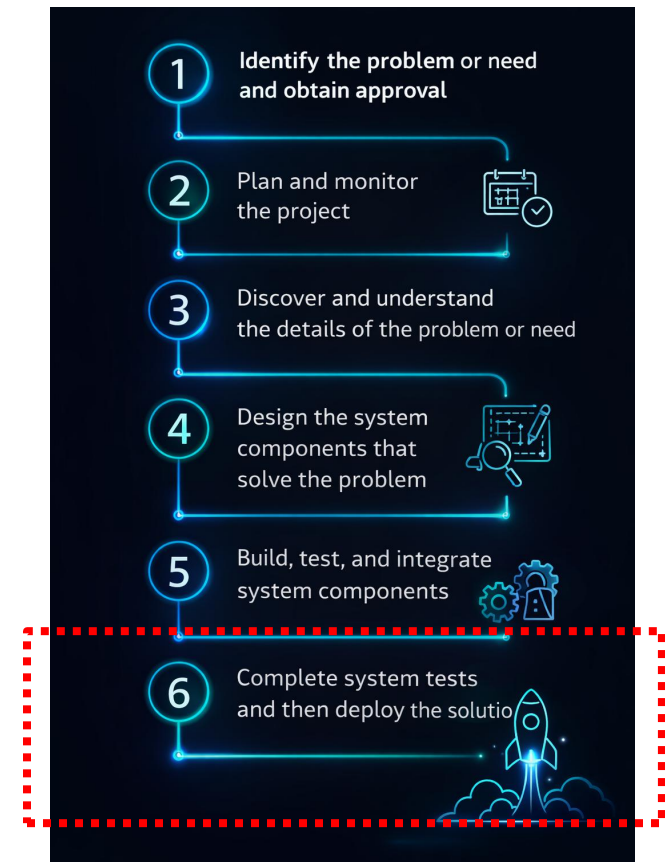
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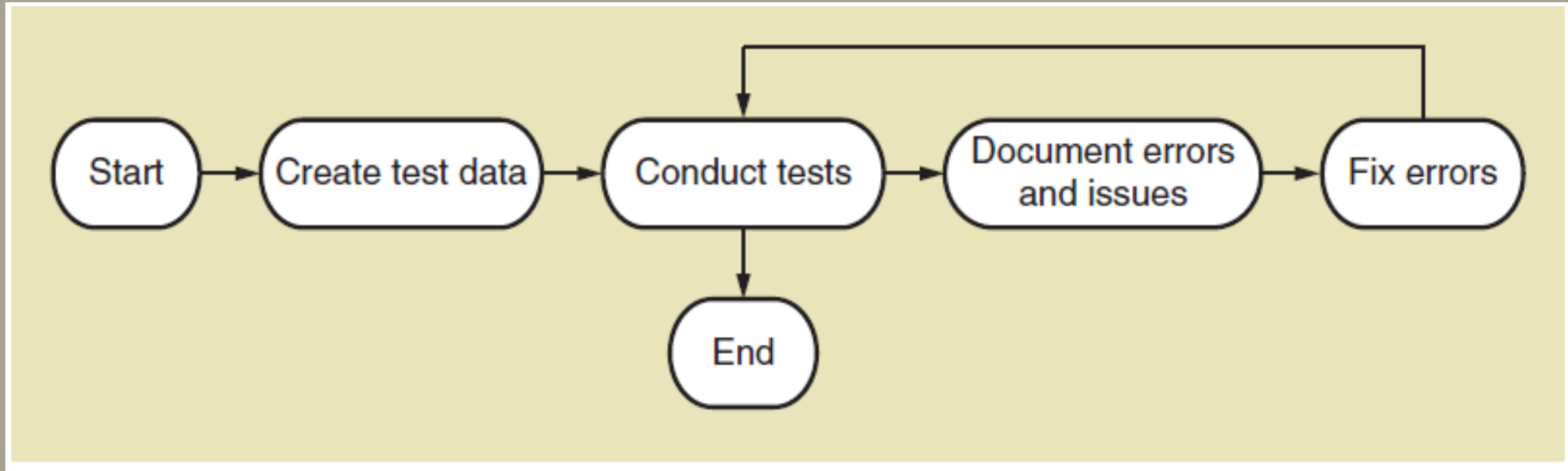
Code Example for One Class

D-6: Activities

Core Process 6: Complete System **Testing** and **Deploy** the System

- Perform system functional testing
- Perform user acceptance testing
- Possibly deploy part of the system





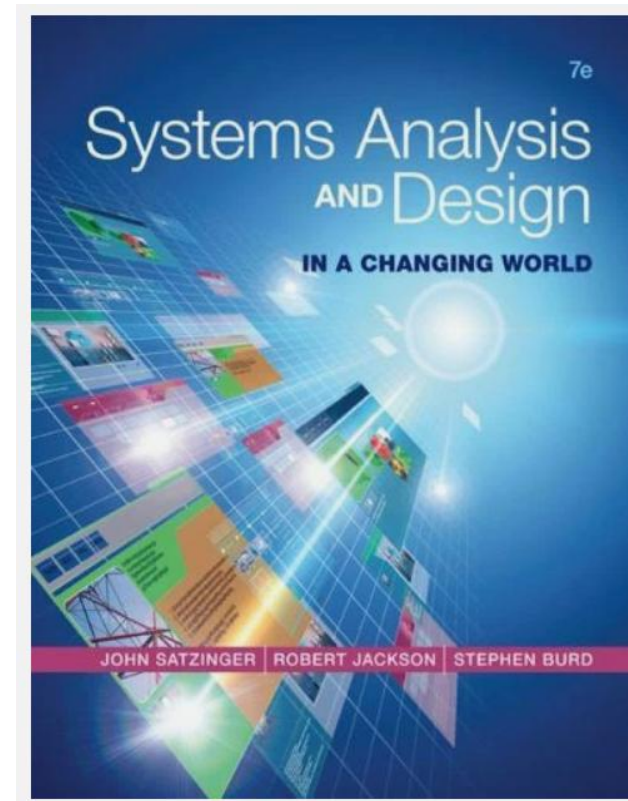
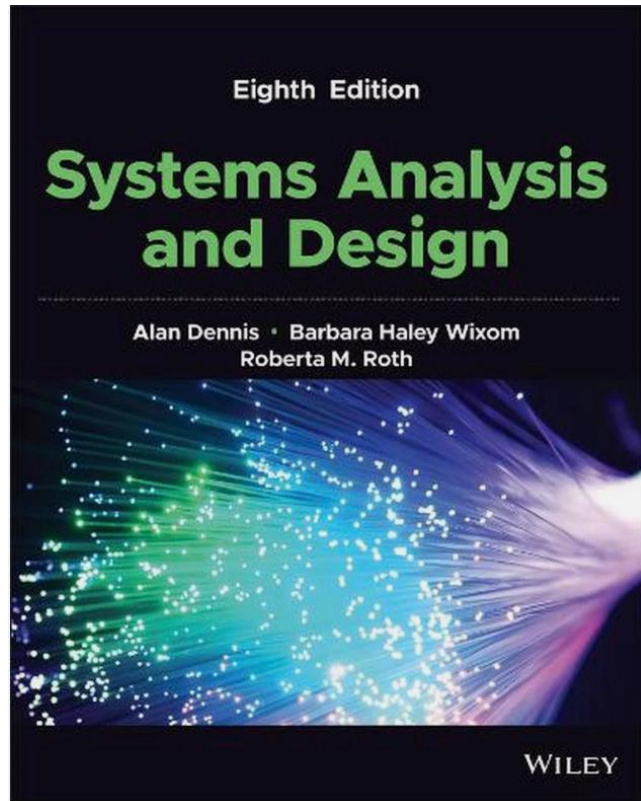
Workflow of Testing Tasks

First Iteration Recap

- This was a 6-day iteration of a small project
 - Most iterations are longer (2 to 4 weeks)
 - This project might be 2 iterations
 - Most projects have many more iterations
- End users need to be involved, particularly on days 1, 2, 3, and 6.
- Days 4 and 5 involved design and programming concurrently.

Computer application **Project**
Systems analysis
Information system development process (methodology) **Information system** **System Design**
Class Diagram **SDLC** Database Schema
Design Class Diagram High Level Structural Design (Architectural Design)
System Development Lifecycle (SDLC)

Summary of Terms



References